

Background:

Options for running Segment Anything Model (SAM):

1. Run on third-party GPU servers like Collab and Kaggle
2. Run on a local computer.

The drawback to the first option is that if you would like to use more resources (memory and processing power) you need to set up a payment plan.

Problems:

The biggest barrier that I am currently encountering is our images that we would like to analyze are very resource intensive and we cannot compile the program without more GPU memory.

In addition, the model itself is very resource intensive and thus I have yet to find a solution to be able to run the program on my computer. *program cannot get past setting up model masks without running out of VRAM*

Potential Solutions:

- Reduce image quality to run in program.
- Change image sizing – git issue suggestions
- Use Fast-SAM instead of SAM
- Use different deep learning segmenting model – https://www.youtube.com/watch?v=J_XSd_u_Yew

Setup for Local SAM environment:

1. Install Cuda based on GPU model.
 - a. Configure conda to communicate with GPU.
2. Install Anaconda and Anaconda explorer.
 - a. Create a new environment in Anaconda explorer with python version 3.
 - b. Open a local Jupiter notebook.
 - c. Run python notebook script in repository.
 - d. Download a local copy of the SAM model and include it in the weights folder within the notebook.

Video Instructions:

Will create this once I know for certain it can work on local computer

SAM Process:

Issues:

<https://github.com/Uminosachi/sd-webui-inpaint-anything/issues/22>

<https://github.com/pytorch-labs/segment-anything-fast/issues/110>

The setup of the masks for this model uses up all the memory of the GPU and terminates the execution. The solution that I am working towards is using FastSAM. Since FastSAM uses a significantly lower number of masks and training data, the GPU that I have might have an easier time running the model. Based on what I've read, FastSAM and SAM work the same way fundamentally.

FastSAM:

Issues:

<https://github.com/JonathanSalwan/Triton/issues/1122>

<https://triton-lang.org/main/getting-started/installation.html>

FastSAM is encountering some dependency issues with triton. This is what I am currently working on to get FastSAM to work.

SAM: